NEXUS-AI PORTAL

Institutional Academic Research & Intellectual Property Management System

Executive System Architecture, Core Capability & AI Innovation Report

1. EXECUTIVE SUMMARY & PROBLEM STATEMENT

Modern higher-education institutions face significant challenges in managing the academic research lifecycle. Capstone projects, student publications, and innovative ideation are typically fragmented across spreadsheets, emails, and disconnected portals. Consequently, high-impact patentable inventions are frequently overlooked or delayed in filing.

Nexus-AI solves this institutional bottleneck by delivering a unified, enterprise-grade digital portal bridging Students, Faculty Mentors, Academic Administration, and Institutional IP Cell Officers. The platform integrates real-time milestone telemetry, AI-driven prior-art novelty scoring, and automated patent filing workflows to maximize institutional IP impact.

2. CORE PLATFORM CAPABILITIES

• **End-to-End Proposal Lifecycle Management:** Standardizes project proposal submissions, PDF upload parsing, structured abstract evaluations, and multi-stage faculty review workflows.

• **AI Prior-Art & Patent Novelty Scoring:** Leverages LLM and cross-referencing algorithms to evaluate project novelty (calculating 90%+ uniqueness index), extract key claims, and perform global patent database collision checks.

• **Weekly Log & Deliverables Telemetry:** Tracks student progress across 16 academic weeks with hours logged, technical challenges, GitHub prototype links, and automated AI activity summaries.

• **Expedited Patent & IP Filing Engine:** Enables IP Cell Officers to evaluate patent-eligible proposals, assign official Indian Patent Office (IPO) application numbers, map jurisdictional zones, and transition status to IP_FILED.

3. FOUR INTEGRATED GOVERNANCE PORTALS

[Portal] Student Research Portal
- Browse and select S5 Capstone topics from approved institutional catalogue.
- Submit weekly progress logs, prototype GitHub links, and demo URLs.
- Track proposal approval stages via real-time Milestone Timelines.

[Portal] Faculty Mentor Portal
- Review student weekly deliverables, assign 1-5 star ratings, and provide guidance notes.
- Evaluate research proposals: Approve, request revisions, or recommend for Patent Filing.
- Monitor student team progress indicators and prototype repositories.

[Portal] Academic Office Portal
- Create and publish institutional project catalogues with domain & tech stack tags.
- Allocate faculty mentors to student research groups across departments.
- Manage central Academic Helpdesk tickets and resolve student/faculty queries.

[Portal] Institutional IP Cell Portal
- Inspect patent-eligible projects flagged as READY_FOR_IP.
- Review AI prior-art analysis reports, non-obviousness criteria, and novel claims.
- Generate official IPO application numbers, record jurisdictions, and submit patent filings.

4. AUTOMATED MILESTONE STATE MACHINE & WORKFLOW

Projects transition through a deterministic state machine managed by automated milestone triggers:

Stage [SUBMITTED]: Student submits research paper/proposal PDF to portal.
Stage [UNDER_REVIEW]: Assigned Faculty Mentor conducts technical review and evaluation.
Stage [APPROVED]: Faculty signs off on paper quality and mathematical formulation.
Stage [READY_FOR_IP]: Project satisfies novelty threshold and is flagged as Patent Eligible.
Stage [IP_FILED]: IP Cell Officer completes IPO form, application number generation, and formal filing.

5. TECHNICAL ARCHITECTURE & STACK

Frontend Layer: React 19, Vite, Lucide Icons, Bootstrap Icons, Custom Design Tokens (#F5F7FC Canvas, #7C3AED Accent)
Backend Framework: Java 17, Spring Boot 3.2, Spring Security, JWT RBAC Authentication, Spring Data JPA
Database Persistence: Embedded H2 / MySQL Relational Database with Transactional Integrity
API Architecture: RESTful HTTP API Endpoints with CORS configuration & Multi-Part File Storage

6. KEY IMPACT & VALUE PROPOSITION FOR PRESENTATIONS

'Accelerated Invention Commercialization: Cuts IP filing preparation time from months to minutes using AI prior-art analysis.
'Unified Institutional Governance: Single source of truth for Students, Mentors, Deans, and IP Officers.
'Zero Loss of IP Assets: Automated detection ensures high-value capstone innovations are identified before graduation.
'Enterprise Design System: Modern, accessible, and high-performance UI tailored for academic excellence.